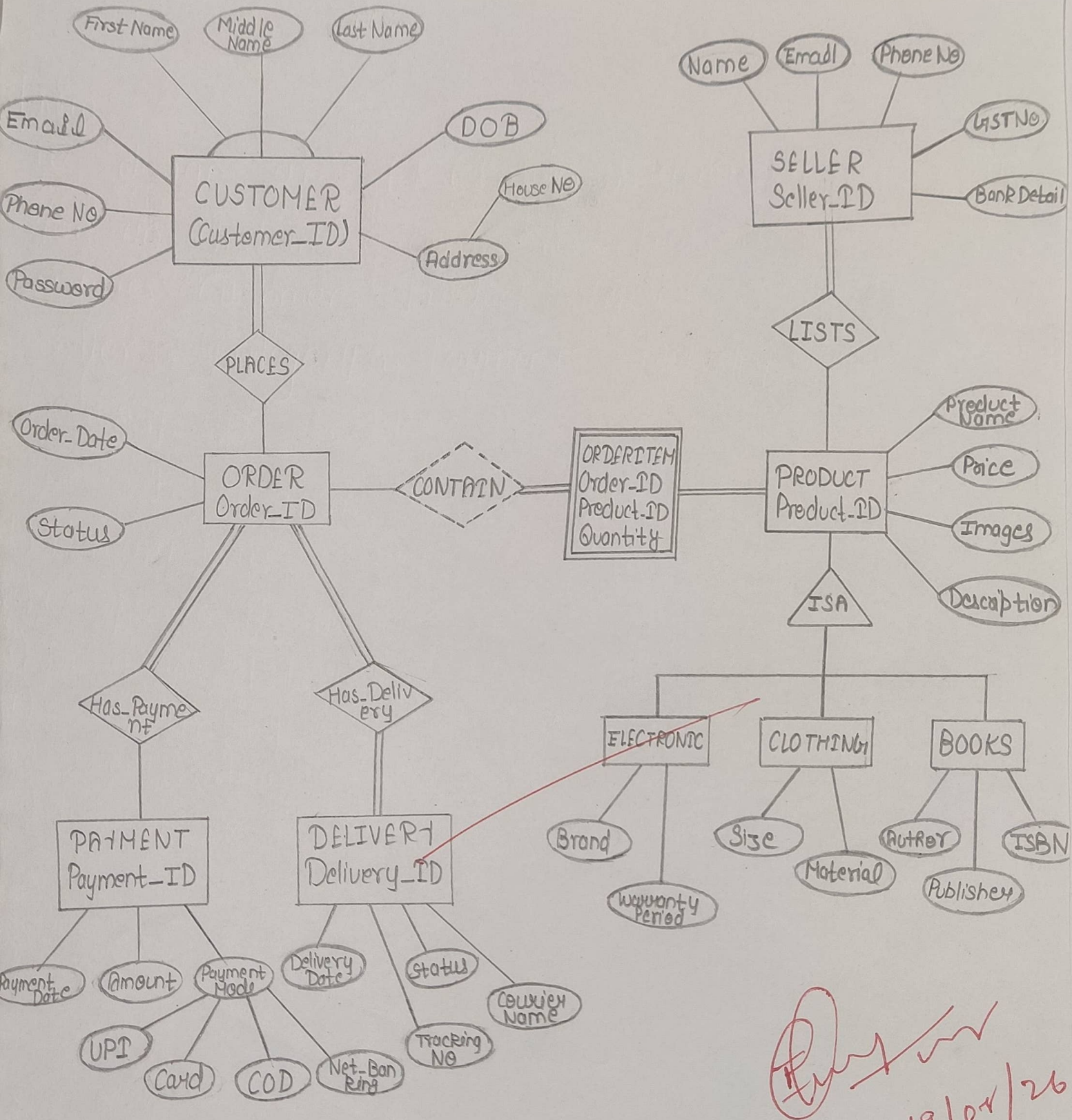
# Experiment - 1

Aim :-

To design an ER (Entity-Relationship) diagram for an Indian e-commerce platform with the entities Customer, Product, Order, OrderItem, Seller, Category, Payment, Delivery and Address.

# ER Diagram for Indian E-Commerce Platform




19/01/26

# Participation Constraints

1) Customer — Order
One customer can place many orders.
Cardinality: 1 : N

2) Order – OrderItem
One order can contain many order items.
Cardinality: 1 : N
OrderItem has total participation.

3) Product – OrderItem
One product can appear in many order items.
Cardinality: 1 : N

4) Seller – Product
One seller can list many products.
Cardinality: 1 : N

5) Order – Payment
each order is associated with one payment.
Cardinality: 1 : 1.

6) Order – Delivery:
Each order is associated with one delivery
Cardinality: 1 : 1.

7) Product – Category
Many products can belong to one category.
Cardinality: N : 1

ER Notations Used.

- Rectangle – Strong entity
- Double Rectangle - Weak entity
- Oval – Attribute
- Double Oval - Multi-valued attribute
- Underlined Attribute - Primary key
- Diamond - Relationship
- Double Line - Total Participation
- Single Line - Partial Participation
- Triangle (ISA) – Specialization
- 1:N - One to Many Relationship.

Output –

The ER diagram for the Indian- E-commerce Platform was successfully designed with the required entities, relationships, primary keys, composite attributes, multi-valued attributes, weak entity, Product Specialization and participation constraints.